

Anomaly detection in UAV-captured crowd images using cumulative frame segmentation and adversarial learning

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ABSTRACT

Anomaly detection in crowds using unmanned aerial vehicle (UAV) captured images is preceded by computer-aided analysis and intelligent learning algorithms. The study is pursued using conventional image processing steps and detection methods. This article introduces a novel anomaly object-detecting method, utilizing the Cumulative Frame Segmentation (AODM-CFS) approach to identify abnormalities in UAV-captured images based on variations in pixel intensity, the data from the VisDrone dataset, and UAV Anomaly Detection. The proposed method segments the maximum intensity varying pixels by examining different pixel occurrences. The cumulative frames are segmented using the maximum repeated intensity pixels to identify objects with maximum feature diversity. The pixel repetition is verified using concatenated adversarial learning, generating repeated and dissimilar pixel maps for various identified frames. These frames are updated using the pixels discovered towards the end of the image. The training for the network map is repeated using segmented frames that rely on maximum feature diversions. Therefore, the abnormal object/ human in the image is identified using the maximum dispersion frame. The proposed method increased detection accuracy by 13.46 %, segmentation precision by 14.08 %, sensitivity by 12.7 %, and specificity by 12.88 %, resulting in an 11.92 % reduction in segmentation error compared to other existing models.

1. Introduction

Detecting crowd irregularities using unmanned aerial vehicle (UAV) images effectively monitors huge public events (Rezaee et al., 2023). UAVs offer extensive aerial views, making it easier to detect unusual crowd behaviours or movements that may indicate potential hazards. A successful anomaly detection system analyses these images to identify abnormalities, such as abnormal population dispersions or movements that could indicate a problem (Zhang, Zong, Shang & Wang, 2023; Ding et al., 2024). Advanced image processing techniques are employed to enhance the accuracy of this detection. The UAV photographs are analyzed to distinguish between typical crowd activities and atypical patterns that may indicate a problem (X. Yang et al., 2024). The approach shortens response times in crucial situations by detecting early symptoms of anomalous behaviour. UAV-based anomaly detection is particularly

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beneficial in situations such as concerts, protests, and festivals, where monitoring large crowds on the ground is challenging. Aerial photos enable authorities to retain a higher situational awareness, resulting in a safer environment (Tang et al., 2024; Gündüz & Işık, 2023).

Analyzing the abnormal behaviour of people in UAV photos is critical for learning how crowds behave in various settings (Yang et al., 2024). UAVs can continuously monitor crowd movement and density to spot abnormalities from normal behaviour, such as sudden rushes or chaotic events. These unusual behaviours are signs of potential safety risks such as crowd surges, violence, or panic (Cui, Liu & Qiao, 2024; Ullah et al., 2022). To process this data, edge detection algorithms classify crowd movements and identify points where behaviour deviates from normal. The analysis helps detect potential crowd safety hazards before they become serious, giving significant insights to decision-makers (Masood et al., 2023; Chen et al., 2023). By recognising these anomalies early, event organisers and security personnel can take proactive steps to prevent calamities. UAV photos offer a unique perspective that enables a more in-depth investigation of crowd dynamics than ground-based monitoring methods. The method provides greater control over crowd behaviour, reducing risks and enhancing public safety (Li, Zhong, Shu & Guo, 2022; Ganokratanaa, Aramvith & Sebe, 2022).

Machine learning (ML) methods are crucial for improving the accuracy of crowd anomaly detection using UAV images. These algorithms evaluate large volumes of visual data and identify complicated patterns that indicate aberrant crowd behaviour (Ullah et al., 2023). ML models may identify and recognize unique patterns more accurately than traditional approaches by utilizing techniques such as convolutional neural networks (Mohamed, Alqahtani, Shalaby & Tolba, 2022). A pre-trained classifier is employed to identify significant features in crowd photos, leading to more reliable anomaly detection (Bamaqa et al., 2022). The incorporation of machine learning (ML) enables the real-time processing of UAV footage, increasing the speed and effectiveness of crowd monitoring. As the models continue to learn from the information, their ability to forecast and detect anomalies improves greatly (Ullah, Ullah, Khan & Baik, 2023; alkato & sakhnini, 2025). ML increases accuracy, decreases false alarms, and ensures that security staff are only alerted to serious hazards. These improvements enhance crowd safety management by enabling faster and more efficient responses to risks (Qaraqe et al., 2024). Conventional anomaly detection models often depend on manually constructed features or standard deep learning methods that do not generalize well across varied and dynamic real-world situations, resulting in elevated false positive rates and diminished resilience.

Furthermore, several current methodologies fail to detect nuanced contextual signals and temporal relationships that are essential for recognizing abnormalities in dynamic crowd behaviours. To address these problems, our proposed AODM-CFS approach utilizes Cumulative Frame Segmentation, which enables the extraction of more comprehensive temporal and spatial characteristics by analyzing frame sequences rather than individual frames. Furthermore, we incorporate adversarial learning to enhance feature representation, enabling the model to differentiate more effectively between typical changes in crowd behaviour and genuine abnormalities.

The primary objective of this study is to present and assess the anomaly object detection method utilizing cumulative frame segmentation (AODM-CFS), which has been customized for crowd images obtained by unmanned aerial vehicles (UAVs). The primary objective is to address the limitations of traditional object identification algorithms, which rely solely on static images and often fail to detect minor irregularities in moving objects. This study aims to enhance the scene's feature representation by integrating temporal information across multiple frames through cumulative frame segmentation. Additionally, by training the model on simulated disturbances, such as illumination changes and occlusions, adversarial learning enhances the model's ability to distinguish between normal and abnormal objects. The goal of this study is to enhance the model's resilience in circumstances such as partial occlusion, fast movement, and other complexity characteristics of UAV-captured imagery while also improving detection accuracy and reducing false positives. The objective is to create a robust system that can accurately identify real-time anomalies and provide highly accurate security monitoring, emergency response, and crowd surveillance capabilities.

The main contributions to the article are:

- A novel Anomaly Object Detection Method was introduced, utilising cumulative frame segmentation to identify abnormalities in UAV-captured images through pixel intensity variations.
- Concatenated adversarial learning was employed to classify and map repeated and dissimilar pixels, thereby enhancing feature extraction and improving the accuracy of anomaly detection.
- The proposed method was validated through extensive ablation studies and comparative assessments, which highlighted improvements in detection accuracy and segmentation precision.
- Experimental results were demonstrated using the VisDrone dataset, which included over 10,000 images and annotated frames, ensuring practical applicability.
- Significant enhancements were achieved in detection accuracy, segmentation precision, and reduced segmentation error, surpassing existing methods such as DeepSDAE, MC-GCAE, and MSMC—Net.

The remainder of the paper is followed by Section 2, which presents a recent literature review of the proposed topic. Section 3 provides a detailed description of the proposed method, utilizing Cumulative Frame Segmentation. Section 4 presents the ablation study, experimental results, and comparative results. Finally, the study's conclusion is presented in Section 5, along with a brief outline of future work.

2. Related works

Yuan et al. (Yuan, Son & Liu, 2022) designed a crowd dynamic modeling approach for multi-resolution aggregation in unmanned

aerial vehicles (UAVs). The designed approach is used to identify the limitations that cause issues for Unmanned Aerial Vehicles (UAVs). The approach implements real-time information to construct relevant models for aggregation. The designed approach enlarges the efficiency and effectiveness level of the systems. Chebil et al. (Chebil, Htiouech & Khemakhem, 2022) proposed a binary linear programming (BLP) model for periodic crowd tracking (PCT) in UAVs. The BLP model is utilized to address PCT issues in UAV-enabled applications. The actual cause and condition of PCT problems are evaluated to provide feasible solutions. The BLP model reduces the computational complexity ratio of the network. The proposed model maximizes the effectiveness range of UAVs.

Zhang et al. (Zhang et al., 2022) introduced a task replacement method for crowd recognition in urban intelligent video systems. The introduced method is utilized in edge-based video systems that require effective processing services. The technique enhances the precision of crowd recognition, thereby securing the safety range of the public. The introduced method improves the performance and minimizes the running time of the process. Wang et al. (Wang, Luo & Zhou, 2024) developed a contrasting pattern estimation approach for anomaly detection in urban crowd flow. The developed approach identifies anomalous or abnormal activities in crowded areas. The approach categorises the pattern according to influencing factors to facilitate further processes. The developed approach expands the overall accuracy and efficiency range of anomaly detection in crowds.

Luo et al. (Luo et al., 2024) proposed a new crowd-level abnormal behavior (CAB) detection method in crowd events. A multi-scale motion consistency network (MSM-Net) is employed to analyze the spatial and temporal features for the detection process. The method calculates the exact behavioural patterns of people in crowded events. It is used to ensure the safety of people at crowd events. Experimental results show that the proposed method increases the precision of CAB detection. Zhang et al. (Zhang et al., 2022) designed an influence-aware attention network for anomaly detection in surveillance video systems. The developed model is used to identify patterns of continuous movement in people via surveillance videos. It is used to address issues caused by anomalies during video recording. The designed model increases the robustness and effectiveness range of the systems.

Yan et al. (Yan, Xiong & She, 2023) proposed a memory clustering autoencoder method for detecting anomalies in human actions in surveillance videos. The technique detects the pseudo-labels and parameters presented in the videos. The proposed method identifies the difference between normal and abnormal actions or movements of people in crowded areas and events. The proposed method elevates the precision level of detection in surveillance videos. An enhanced version of (Yan, Xiong & She, 2023) is introduced by Li et al. (Li et al., 2021), which incorporates motion consistency. The introduced model is known as a variational abnormal behaviour detection (VABD) model in the context of surveillance systems. The actual motion and movement of the people are calculated to detect behavioural patterns. The model discriminates between abnormal videos captured via surveillance cameras. Compared with others, the introduced model increases detection accuracy and enhances performance.

Li et al. (Li et al., 2022) developed a new attention-based anomaly detection method for multi-view surveillance videos. The developed method detects the types and classes of anomalies in the given input videos. The method addresses the issues by utilizing modules that reduce the process response time. It also predicts the cause of action based on the crowd's behavioural data. The developed method maximizes the effectiveness range of the systems. Castellano et al. (Castellano, Cotardo, Mencar & Vessio, 2023) designed a density-based clustering technique for crowd flow detection from drones. A fully convolutional neural network (FCNN) algorithm is implemented to detect the centroids from the crowd. The actual flow of the crowd and abnormal activities are identified from drone videos. The designed technique increases the accuracy level of the systems.

An improved version of (Li et al., 2022) is proposed by Amin et al. (Amin et al., 2023) using a quantum convolutional neural network (QCNN). CNN and quantum computing technology are combined to detect anomalies in crowded areas. The behavioural patterns and factors of the people are identified to estimate the type of anomalies. Experimental results show that the proposed model improves the accuracy and efficiency range of the detection process. Ou et al. (Ou, Zhu, Chen & Liu, 2023) developed a new human behaviour recognition model using a neural network for video surveillance systems. The developed model ensures users' overall safety and security range. Abnormal behavioural patterns and activities are identified through video surveillance. The developed model enhances the performance and effectiveness level of the recognition in surveillance systems.

Li et al. (Li et al., 2023) introduced a new video anomaly detection based on scene classification. The introduced detection method utilises a generative adversarial network (GAN) to detect anomalies in video. A k-means clustering algorithm is also employed here to categorize types of anomalies based on their severity. The method minimizes the computational cost of the anomaly detection process. The introduced method increases the performance range of the systems. Zhong et al. (Zhong et al., 2024) developed a unifying framework for dense crowd counting. A Gaussian kernel filter is used here to filter the video as a necessity. The developed framework analyzes the head maps and binary feature maps, producing optimal data for counting. It also eliminates latency during the counting process. Compared to others, the developed framework maximizes the accuracy of head counting in crowds.

Yang et al. (Yang et al., 2023) proposed a deep neural model for crowd anomaly detection in videos. The proposed model uses a stacked denoising auto-encoder that identifies anomalies in input videos. The model eliminates unwanted noise from the videos, thereby enhancing the feasibility range of the detection process. The proposed model maximizes the overall performance and efficiency level of the systems. Asl et al. (Asl, Karasfi, Masoumi & Keyvanpour, 2024) developed an abnormal behavioural recognition method in crowded environments. The developed method uses a semi-supervised deep learning (DL) and hierarchical approach for behaviour recognition. The technique detects abnormal behavioural patterns in overcrowded areas to facilitate further processing and gather relevant data. The developed method increases the significance and reliability range of crowded environments.

Ziao Wang et al. (Wang, Chen & Chen, 2024) proposed the Skeleton-based Iterative Perspective Optimization Algorithm for UAV Patrol Tracking of Large-Scale Pedestrian Abnormal Behavior. An algorithm based on iterative learning control is suggested for tracking huge crowds using information about the human torso and shoulder skeleton, and a PID iterative learning control algorithm is developed to enhance the control effect when pedestrian motion is approximately repeatable, both of which aim to address the problem of recognizability in images captured during UAV patrol. Both the number of pedestrians photographed and the duration of

their activities that can be continuously observed can be improved using this technique. Next, experimental findings confirm the algorithm's efficacy and usability. In embodied intelligence, the proposed approach will facilitate the study of interactive perceptual technologies.

Niu, Zuodong, et al. (Niu, Pi, Jing & Liu, 2024) suggested the Partial Information Interaction Convolution (PIIConv) with Global Collaborative Fusion (GCFusion) for Crowd Counting and Localization in UAV RGB-T Images. According to the empirical results, the newly proposed network improves inference time across several platforms, drastically decreases floating-point operations (FLOPs), and achieves competitive results on the DroneRGBT dataset. This research is crucial for real-time UAV deployment, providing a computationally efficient framework for RGB-T crowd counting that strikes a good balance between accuracy and resource efficiency.

Murat Bakirci (Bakirci, 2025) recommended the use of remote sensing and deep learning for Vehicular mobility monitoring, utilizing a UAV-based mobile computing platform. An improved mobile computing platform for real-time vehicle recognition and traffic monitoring is achieved by upgrading the hardware and software of a test unmanned aerial vehicle (UAV). The UAV's enhanced communication infrastructure enables quick data transfer to control stations, facilitating prompt decision-making and informed action. Several YOLOv8 size variations were suitable for real-time applications when evaluated on the Jetson Nano platform. The YOLOv8n version stood out, attaining quicker detection times and an accuracy rate of 0.706 in vehicle identification tests.

Additionally, YOLOv8n was tested against several versions of the cutting-edge algorithms YOLOv9, YOLOv10, and YOLOv11. It

Table 1

Summary of the existing models.

Author(s)	Methodology	Results	Advantages	Limitations
Yuan et al. (Yuan, Son & Liu, 2022)	Crowd dynamic modeling approach for aggregation of multi-resolution in UAV	Enhances the efficiency and effectiveness of UAV-based crowd modeling	Real-time information improves model accuracy	Limited adaptability to varying environmental conditions
Chebil et al. (Chebil, Htiouech & Khemakhem, 2022)	Binary Linear Programming (BLP) for Periodic Crowd Tracking (PCT) in UAVs	Reduces computational complexity and improves UAV effectiveness	Optimizes PCT problems efficiently	Not evaluated in real-world, large-scale crowd scenarios
Zhang et al. (G. Zhang et al., 2022)	Task replacement method for crowd recognition in intelligent video systems	Improves the precision of crowd recognition and public safety	Enhances performance and reduces processing time	Limited scalability for large surveillance networks
Wang et al. (Wang, Luo & Zhou, 2024)	Contrasting pattern estimation approach for anomaly detection in urban crowd flow	Improves accuracy and efficiency in detecting anomalies	Categories patterns based on influencing factors	Requires extensive labelled data for high accuracy
Luo et al. (Luo et al., 2024)	Multi-Scale Motion Consistency Network (MSM-Net) for abnormal behaviour detection	Increases precision in detecting crowd-level abnormal behavior	Effective spatial-temporal analysis of motion	Computationally expensive for real-time deployment
Zhang et al. (S. Zhang et al., 2022)	Influence-aware attention network for anomaly detection in surveillance videos	Enhances the robustness and effectiveness of surveillance systems	Identifies continuous movement patterns	Requires high-quality surveillance video input
Yan et al. (Yan, Xiong & She, 2023)	Memory clustering autoencoder for human action anomaly detection	Improves accuracy in detecting human actions in videos	Distinguishes normal and abnormal actions efficiently	Sensitive to parameter tuning and pseudo-labels
Li et al. (Li et al., 2021)	Variational abnormal behavior detection (VABD)	Enhances anomaly detection accuracy in surveillance systems	Improves motion consistency evaluation	Requires large training datasets for optimization
Li et al. (Li et al., 2022)	Attention-based anomaly detection in multi-view surveillance videos	Improves process response time and anomaly classification	Reduces computational cost	Limited adaptability to non-structured crowd behavior
Castellano et al. (Castellano, Cotardo, Mencar & Vessio, 2023)	Density-based clustering technique with FCNN for crowd flow detection	Increases the accuracy of crowd flow analysis in drone videos	Identifies crowd centroids effectively	It does not consider real-time adaptability for different crowd densities
Amin et al. (Amin et al., 2023)	Quantum Convolutional Neural Network (QCNN) for anomaly detection	Enhances accuracy and efficiency in crowded area anomaly detection	Combines quantum computing with CNN for improved results	High computational cost and complex implementation
Ou et al. (Ou, Zhu, Chen & Liu, 2023)	Neural network-based human behavior recognition for surveillance	Improves safety and security via behavioural analysis	Detects abnormal patterns effectively	Lacks generalization across diverse surveillance settings
Li et al. (Li et al., 2023)	Video anomaly detection using GAN and k-means clustering	Improves anomaly classification accuracy	Reduces the computational cost of detection	GAN-based models require extensive training time
Zhong et al. (Zhong et al., 2024)	Unifying framework for dense crowd counting using Gaussian kernel filtering	Enhances accuracy in head counting from videos	Eliminates latency in the counting process	Performance degrades in low-light conditions
Yang et al. (Yang et al., 2023)	Stacked denoising autoencoder for crowd anomaly detection	Improves efficiency by removing noise in surveillance videos	Enhances the feasibility of anomaly detection	Requires extensive computational resources for real-time use
Asl et al. (Asl, Karasfi, Masoumi & Keyvanpour, 2024)	Semi-supervised deep learning for abnormal behavior recognition	Enhances reliability and significance in crowded environments	The hierarchical approach improves recognition accuracy.	Requires large-scale labelled datasets for training.

showed the optimal trade-off between recall and accuracy, and although it fell short of the top spot in terms of precision, the gap was small. The technology accurately tracks designated vehicles and reports their trajectories to the command center in real-time.

Jeong Do Yoo et al. (Do Yoo et al., 2025) discussed memorising normality for UAV anomaly detection using a few sensor values. This research uses preprocessing techniques, including feature selection, timestamp pooling, and missing value imputation, to produce a condensed feature set. The author then employed an autoencoder variant with a memory module, which successfully detects anomalies. This variant stores archetypal benign patterns after learning. Therefore, when examined with two datasets (ALFA and UA), MeNU outperforms the prior research with AUC values of 0.9856 and 0.9988, respectively. Khosravi, M. R. et al. (Khosravi et al., 2023) addresses real-time emotion recognition in dynamic crowd scenarios to improve human-vehicle interaction safety. It presents a hybrid model for emotion classification, utilising modified transfer learning for feature extraction and fuzzy logic ranking to address ambiguity. The method accurately detects and ranks crowd emotions from surveillance data, proving robust in complicated urban environments. However, its efficacy depends on pre-trained models and may be limited when generalizing across crowd behaviours and environmental variables. Tomar Ankit et al. (2024) (Ankit, Santosh & Bhasker, 2024) demonstrate PeopleNet, a framework based on deep learning developed for accurately counting people in dynamic environments using head-mounted movement cameras. To address issues such as motion blur, occlusion, and perspective distortion, it combines a lightweight backbone with a temporal aggregation module. To maintain its resilience across a wide range of movement speeds and densities, the model incorporates contextual and spatial factors. Conventional detection-based counting approaches are outperformed by this method, which displays outstanding performance on benchmark datasets.

Nevertheless, the fact that it is dependent on motion patterns restricts its capabilities for use in static surveillance installations. Hussain, A. et al. 2024 (Hussain et al., 2024) Models spatial and temporal connections with a dual-stream architecture and Transformer modules to improve video anomaly detection. Attention mechanisms blend appearance and motion features from the two streams to emphasize abnormal places. Transformer layers enable the network to capture long-range dependencies within video frames. The technique improves detection accuracy and performs well on numerous benchmark datasets. The computational complexity may limit its use in real-time or edge-based surveillance systems. Table 1 shows a summary of the existing models.

Conventional, manually operated feature-based approaches are inadequate in real-world settings because they rely on statistical models and predetermined motion patterns, which render them very vulnerable to changes in illumination, occlusion, and pedestrian density. While deep learning-based methods provide better performance, they often employ static frame-based analysis, which fails to capture critical temporal correlations in crowd movements, resulting in less-than-ideal anomaly identification in real-time situations such as disaster management, anomalous vehicle behaviour, border and perimeter security, military bases, airports, national borders, and maritime surveillance. Modern models that incorporate attention mechanisms or recurrent networks to enhance feature representation often struggle to differentiate between typical population changes and genuine abnormalities, resulting in numerous false positives. Likewise, many current approaches aren't efficient or scalable enough for real-time UAV monitoring applications. This study presents the AODM-CFS approach, which incorporates Cumulative Frame Segmentation, to address these shortcomings. This technique enhances temporal feature extraction by utilizing a sequence of successive frames rather than individual images, thereby providing a more comprehensive representation of crowd dynamics. In addition, this study utilizes adversarial learning to enhance feature discrimination, thereby reducing the misclassification of typical actions as abnormalities. The detection accuracy, resilience, and adaptability to complex settings are all significantly improved by this combination, making our technology more suitable for real-time UAV surveillance applications.

3. Proposed method using cumulative frame segmentation

Anomaly detection in crowd images is achieved using an intelligent learning algorithm. By leveraging UAV images, a maximum dispersion frame is identified where the abnormalities are detected. Here, the UAV features are extracted, the intensity of the input image is estimated, and anomaly detection is performed. The necessary features, such as region and edge, are extracted from the captured UAV image. The intensity is identified from the extracted region, such as a grayscale or coloured image. The proposed method is given in Fig. 1a.

The computation is performed based on pixel-based intensity detection and provides a result indicating where abnormalities are detected in the crowd image. This detection indicates that the concatenated image is based on intensity, meaning the starting and ending pixels are in the same position. Here, the old and new pixels are combined into a single-pixel image, considering that this anomaly detection is complete. Here, this method aims to identify anomalies in crowded images and ensure accuracy. The segmentation is evaluated from the frame region where the concatenated images are mapped. The intensity estimation is classified as repetition and dissimilarity, using concatenated adversarial learning to identify various frames. Based on this learning, anomaly detection is performed, where the preliminary equation serves as the objective, which is based on the intensity of the pixel and is presented below.

$$\partial a' = \sum_{UA'}^{Iy'} Fe_x + (reg' + D_\sigma) - AN' \quad (1)$$

The objective is equated in the above equation; the evaluation is represented as $\partial a'$ where UAV is the input image. The feature extraction is described as Fe_x , reg' is labelled as a region, the detection is represented as D_σ . The anomaly detection is done from the extracted featured region, and the anomaly is described as AN' . The intensity is described as Iy' and based on this, the concatenated process is evaluated. Here, the maximum dispersion frame is identified by introducing the CAN method. The necessary feature is

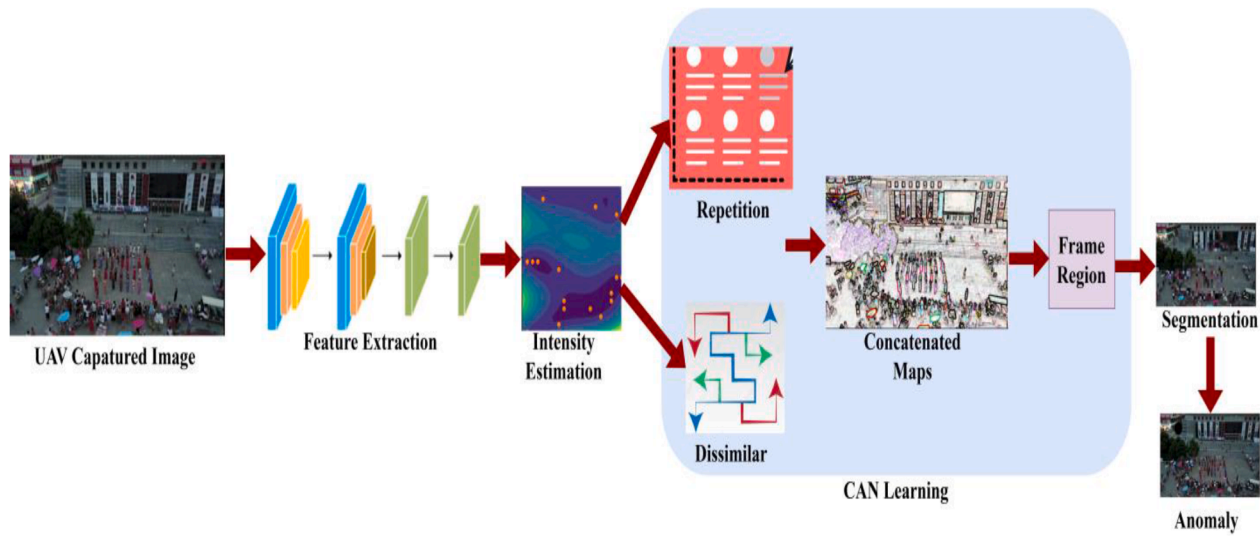


Fig. 1a. Proposed AOD method using CFS.

extracted from the captured UAV image, and the intensity of the colour or grayscale image is evaluated. From the intensity image, the region is extracted to detect whether any abnormalities are present.

Fig. 1b illustrates how the proposed Anomaly-Oriented Detection Method with Cumulative Frame Segmentation (AODM-CFS) operates for crowd anomaly investigation using UAVs. The first stage in the pipeline involves preprocessing the video streams captured by UAVs. This includes extracting frames, converting them to grayscale, and reducing noise. Next is cumulative frame segmentation, which utilizes pixel-wise difference analysis to maintain synchronization over time. Feature extraction and intensity estimation processes help to identify edges and select regions of interest (ROI). A Concatenated Adversarial Learning module, featuring a generator-discriminator pair and ROI-based segmentation, takes these segmented frames as input. The learned features aid in anomaly categorization by determining anomaly scores. After classifying, the algorithm examines patterns of frame-level repetition and dissimilarity to identify anomalies in the correct locations, ensuring that AODM-CFS accurately detects anomalous behaviors in dense crowds.

Based on this detection, the anomaly is detected using the concatenated mapping under CAN learning. This segmentation helps to classify the intensity, including repetition and dissimilarity, of the image. From this process, detection is achieved at the pixel level of the image, which varies according to the region of interest. The area acts as one of the features in a UAV image and, based on this anomalous behavior, is detected in the crowd scene. The feature extraction is processed in the following equation after this objective evaluation. The variable T_1 captures region-intensity anomaly interactions, T_2 adds dispersion and UAV-based adjustments, T_3 accounts for pixel-wise intensity variation and smoothing, with the final mc_r normalizes or offsets crowd-based bias.

$$Fe_x = \begin{cases} T_1 + T_2 + T_3 - mc_r \\ T_1 = reg' * \prod_{UA'}^{AN'} D_\sigma + (AL_y * reg') - xi_p + \left\{ [(xi_p + reg') * Sel] + \prod_{sm_0} Iy' \right\} * AN \\ T_2 = D_\sigma(UA' + Iy') * |reg' + mc_r| \\ T_3 = Iy' + (D_\sigma * AN) - UA' + (sm_0 + AN(Iy')) \bullet D_\sigma \end{cases} \quad (2)$$

The feature extraction is computed from the captured UAV image, where it considers the region and pixel, and the pixel is represented as xi_p . The selection of region is described as Sel , where sm_0 is the estimation, mc_r is the crowd image. The evaluation is done for the region-based pixel distributed on the UAV image. The analysis is symbolized as AL_y , where the intensity is estimated from the region. The extraction is the necessary region to determine whether a similar pixel matches the prior dataset. The result is generated as the anomaly is detected, with intensity estimation.

The intensity estimation is considered for the input image, where the pixel value of each region is examined, and the value at which the variation occurs is identified. The value is represented as the brightness and colour of each pixel in the image. The value ranges from 0 to 255; by considering this value, the pixel-based region is extracted, and the result is produced. From this computation, the region-based feature is extracted $(AL_y * reg') - xi_p$. Here, intensity estimation is done from the crowd image $((sm_0 + AN(Iy')) * D_\sigma)$.

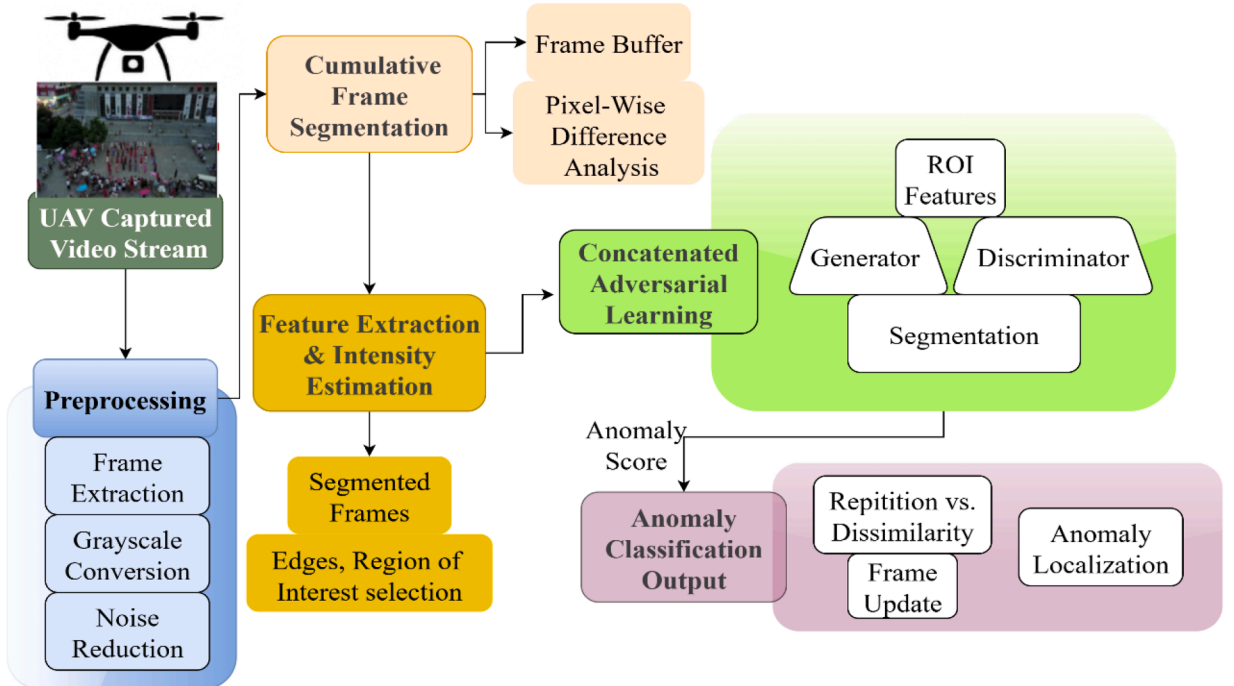


Fig. 1b. High-level architecture of the AODM-CFS framework for anomaly detection.

Here, the intensity estimation is derived from the equation below.

$$\begin{aligned}
 Iy'(sm_0) &= Fe_x(UA' * Con) \\
 &+ \sum_{reg'} xi_p * [(Sel + UA') * mc_r] \\
 &+ |Sel + reg'| * Fe_x \\
 &+ (AN' * D_\sigma) \\
 &+ Fe_x
 \end{aligned} \tag{3}$$

The intensity estimation is the light reflected from the corresponding object/particular. In common words, an image is a two-dimensional function that coordinates with x,y axes. In some input images, the intensity is either coloured or grayscale, allowing the required part to be detected. Here, for better understanding, the intensity is detected in the region of interest, and from this, the pixel is identified to detect anomalous behavior in the crowd image. The evaluation takes place by using the image intensity where the concatenated image is considered, and it is described as *Con*. From the required region, the pixel is extracted as a region of interest and identified by its intensity value. The intensity estimation flow diagram is given in Fig. 2.

The Iy' evaluation is performed to identify the same and different intensity features using Fe_x . If the $xi_p \in Iy'$ that is similar for any mc_r input, the (x,y) intends are common. If a common $xi_{p_{is}}$ identified, then Fe_x is halted until a D_σ is observed. Therefore the AL_y takes place for the new Fe_x identified where it is true, and this means there is some variation in xi_p other than *Sel*. Thus, if $Iy' = \sum xi_p$ then, the $Iy'(sm_0)$ is monotonous for identifying reg' . These reg' are grouped to form a frame where *con* is false. Therefore, the feasibility of *con* or *Sel* is differentiated from this Iy' evaluation as presented in Fig. 2. The intensity value is generated based on the two axes, and according to the image parameter, the estimation is processed. Here, the image parameter for intensity is location, and anomaly detection is evaluated based on the location of the crowd image. Thus, the selection of the region is done, and from this feature, extraction is processed $|Sel+reg'| * Fe_x$ and this is how the intensity estimation is computed in the above equation. The post to this concatenated image is analysed by introducing CAN learning.

3.1. CAN learning

The concatenated image refers to the pixel start and end positions, where the old and new pixels are joined together. The pixel repetition is verified using concatenated adversarial learning, which generates repeated and dissimilar pixels for various frames. The adversary's learning is misguiding the model with false input, which is detected and avoided to enhance accuracy. This approach is used to improve the accuracy and performance of image intensity among the UAVs and to identify anomalous behaviour. The equation below is used to evaluate the concatenated image.

$$\begin{aligned}
 \partial a' &= UA'(Fe_x) + sm_0 * (mc_r + Sel) * reg' + (reg' * UA') * \left(AN' + \frac{AL_y}{Iy'} \right) + sm_0(UA') + Sel(Fe_x - D_\sigma) + (D_\sigma(UA')) * (AL_y + Con) + xi_p \\
 &* (D_\sigma + UA' * Fe_x) + Iy'
 \end{aligned} \tag{4}$$

Anomaly score thresholds are dynamically determined from the feature-extracted regions, through a data-driven approach where the anomaly is denoted as AN' . The anomaly score $\partial a'$ defined in Eq. (4), aggregates several factors, including the UAV-based feature projection $UA'(Fe_x)$, image density Iy' , region level variations reg' and statistical dispersion D_σ . Rather than using a fixed threshold, establish a context-sensitive threshold based on the difference between Re_p and Dis' components from Eqs. (5)–(7), which reflect how similar or different a region is across concatenated frames. The dispersion factor D_σ is used to modulate this threshold, adapting it per frame based on crowd density and pixel variability. Thus a region is classified as anomalous when $\partial a'$ exceeds this learned, adaptive threshold, making the anomaly detection score more robust to environmental noise and variations in the UAV-captured scene. This adaptive method has been applied throughout all experimental evaluations, ensuring that detection performance has not biased by fixed parameters, but rather optimized per scene based on the observed anomaly score.

The evaluation for the concatenated image is done on the above equation, and it is labelled as $\partial a'$. Here, the detection occurs from the crowd image and provides accurate feature extraction. From the crowded scene, the abnormalities are detected by locating the position. The location is an image parameter. Locating the position is the region of interest where the pixel-based analysis is evaluated. Here, analysis is done with a concatenated image by detecting the anomalous behaviour, and it is represented as $[(D_\sigma(UA')) * (AL_y + Con)]$ and this is how the evaluation is processed under the CAN learning framework.

If any misleading process occurs, it results in incorrect predictions and inaccurate detection for this evaluation. The training is used to improve the output by providing the proper prediction with an accurate value. The input is acquired as a crowd image, and then intensity estimation is performed with classification to identify anomalous behaviour; this is how machine learning works, providing a predicted value based on the intensity estimation. The classification is processed in the equation below.

$$L\Delta' = \frac{(mc_r + D_\sigma) * \sum_{Iy'} sm_0 + (xi_p * UA') + |AN' * D_\sigma| + \partial a' * (Con + UA') * AL_y - xi_p + (Sel * UA') + Iy'}{Re_p} \cdot \frac{Dis'}{Dis'} \tag{5}$$

The classification is represented as $L\Delta'$, and the repetition is described as Re_p , Dis' is symbolized as dissimilar. The first derivative refers to repetition, where the pixel-based estimation is done to detect the anomaly from UA' . Here, the image is repeatedly used to

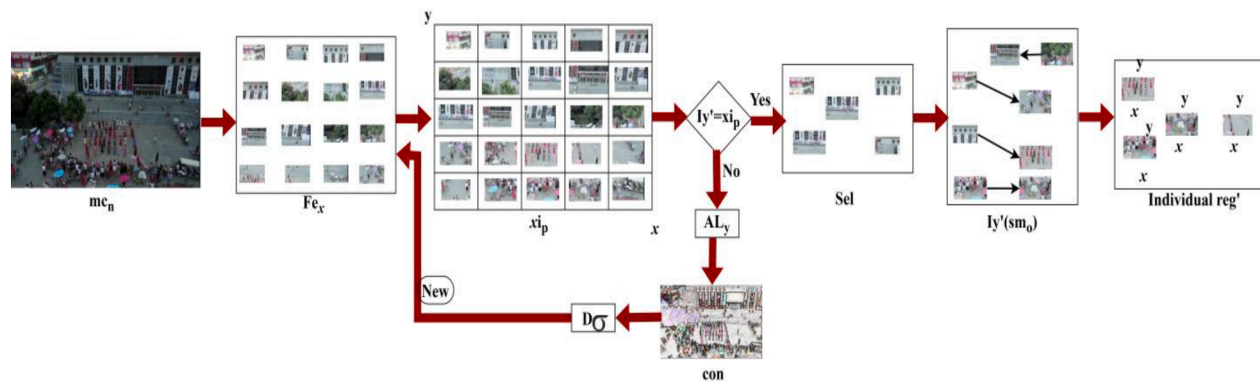


Fig. 2. Intensity estimation flow.

detect abnormalities, whereas the second expression represents a dissimilar process where pixel selection is evaluated from a fixed region of the image. Based on this classification, the intensity estimation is computed for the repetition and dissimilarity of the image, as formulated below.

$$Re_p = Con(UA') + Fe_x * \left(\frac{AL_y + mc_r}{\sum_{ly} sm_0} \right) \quad (6)$$

The repetition is computed in the above equation. Here, the analysis is performed on crowd scenes, and the intensity is estimated in either colour or grayscale. The repetition occurs due to a location change; in this case, the repetition is evaluated based on the extracted pixel-based region. Post to these dissimilarities is computed below.

$$Dis' = AN(D_\sigma) + \left(\prod_{m_{c_r}} reg' + Fe_x \middle/ (x_{i_p} * UA') \right) * TR_\alpha \quad (7)$$

The dissimilarities are detected when no match is found with the current image. If any dissimilarities occur, the training is given accordingly, and it is labelled as Tr_n . It is computed using CAN learning, where the pixel-based estimation is processed to avoid dissimilar images and detects the anomaly from UA' . The repetition is mapped with the concatenated image, which is derived below.

$$mp' = \underbrace{(o' - w_n) * UA'}_{\text{motion shift}} + \underbrace{(xi_p + Re_p) * reg'}_{\text{region-pixel mapping}} + \underbrace{\left(\frac{AL_y + mc_r}{\sum ly sm_0} \right)}_{\text{density normalization}} + \underbrace{(xi_p * UA')}_{\text{pixel-weighted UAV influence}} \quad (8)$$

$$+ \underbrace{[AN * D_\sigma]}_{\text{anomaly dispersion}} + \underbrace{\partial a' * (D_\sigma - Iy' + Fe_x)}_{\text{anomaly score adj}} + \underbrace{(xi_p + D_\sigma) * Sel(reg')}_{{\text{region selection by dispersion}}$$

The mapping is done on the concatenated image, and it is labelled as mp' , which indicates the region selection. The region selection is evaluated with the pixel and map with the old image, and it is o' ; the current image is w_n . This mapping is done to improve the repetition of concatenated images and identify particular behaviours from the data collection. The mapping with the old image ensures the accurate value based on the pixel by locating the region. The $L\Delta'$ and mp' processes are described in Fig. 3.

The LA' and mp' processes are detailed in Fig. 3 above. This process requires Sel and Con inputs for LA' and sm_o verification. If $D_\sigma = con$, then Re_p is true; the first mp' element is this output. The con output for maximum sm_o ensures reg' differentiation and its association with Dis' is used for mp' . Therefore, the LA' process's Dis' and sm_o process's reg' outputs are conjoined to perform mp' . This mp' is used for Sel and con such that $(x_{ip} + Re_p)$ and $(x_{ip} * UA')$ are the mapping instances. Thus $Sel(reg')$ is useful in training $[AN * D_\sigma]$ and $[Fe_x]$ for LA' and Sm_o verifications. If Fe_x is true, then the number of possibilities for segmentation is high and, therefore D_o is reliable across performance improvements.

3.2. Frame detection and update

The detection is processed from the crowd image and searches for the intensity estimation. Here, the particular frame is acquired from the UAV image and identifies the intensity of the pixel image. The computation is performed on the concatenated image, improving performance and accuracy. The estimation is processed for the number of pixels where the similarities and dissimilarities are detected, making the task easier for anomaly detection. From the concatenated mapping, frame detection is performed, where

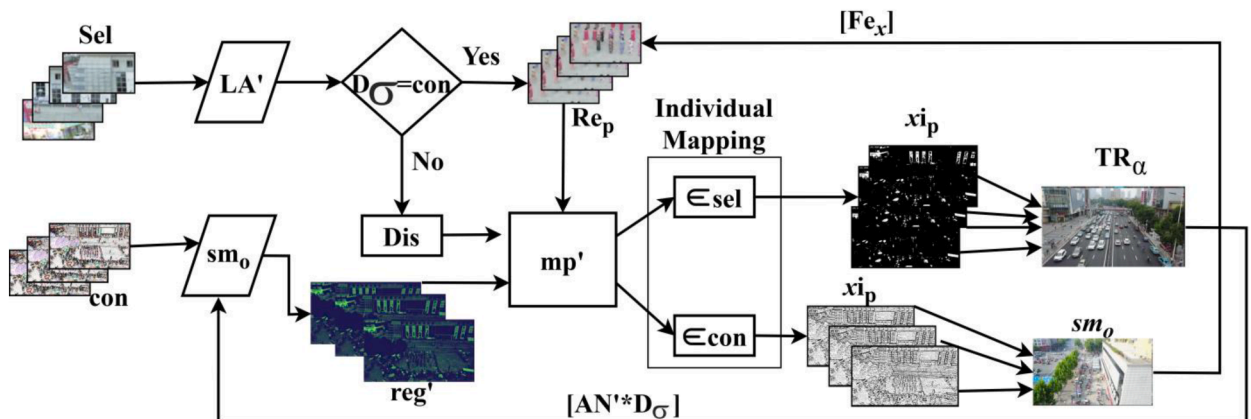


Fig. 3. $L\Delta'$ and mp' processes.

pixel-based region extraction is processed. The frame region is detected in the equation below.

$$D_{\sigma} = \sum_{UA'}^{Fe_x} Iy' + (sm_0 * (Re_p - Dis')) * \{ [(mc_r + AN') + (reg'(xi_p) * AL_y)] + L\Delta' \} \quad (9)$$

The above equation detects the region from which the input is taken in the crowd scene and identifies abnormalities. The evaluation is processed for the concatenated mapping with the old and new images, where the frame-based region is extracted. From the input image, frame splitting is performed, and the first frame is used to compute the region extraction. The region is based on pixel-to-pixel comparison, mapping the first-pixel frame with the old data and checking whether it is abnormal. The detection is evaluated based on the concatenated mapping.

The update is computed once any intensity variation occurs; the frame region is updated if it shows intensity differences. The feature extraction is performed on the acquired frame region for pixel-based intensity. The update UP' is used to improve the accuracy of detection from the gathering. This frame is represented as Fr_m which is updated using the pixels discovered towards the end of the image size. Thus, the frame region is detected with the mapping process, and from this frame update, it is computed and equated below as follows.

$$Fr_m(UP') = D_{\sigma} + (o' - mc_r) * \left(\sum_{Sel} reg' / (sm_0 + Iy') \right) * \prod_{Sel} |xi_p + Fe_x| - Con(UA') \quad (10)$$

The update is done for the pixel where the location is considered and found for the region. Pixel-oriented detection is performed for the frame region, considering the area within the frame. If any possibilities are identified, a frame update is done for further detection. Thus, for every computation, the frame update is utilized; after this, training is provided for the dissimilar image, ensuring maximum feature diversity. Algorithm 1 shows the anomaly detection using the proposed AODM-CFS.

3.3. Training

The training for the network map is repeated using segmented frames that rely on maximum feature diversions. The computation is performed to enhance the performance of the extracted features and parameters. The diversion features are obtained from the training process and are equated in the equation below.

$$TR_{\alpha} = UA'(D_{\sigma} + mp') - o' + \frac{D_{\sigma}}{fm_p} * Fr_m \quad (11)$$

The training is provided for the segmented frames to enhance feature diversification through frame updates. The frame update is processed with the selection process, and mapping takes place; it is described as mp' , and the performance is represented as fm_p . The mapping is performed using the old image and yields an output that relies on diversion, resulting in higher accuracy. And this is how

Algorithm 1

Anomaly detection using proposed AODM-CFS.

Input: UAV-captured video stream **Output:** Detected anomalies **Step 1: Preprocessing**

- 1.%2 Convert each frame to grayscale for feature extraction
- 2.%2 Perform noise reduction and image enhancement (Gaussian filtering)
- 3.%2 Normalize pixel values

Step 2: Cumulative Frame Segmentation (CFS)

- 2.1. Initialize the cumulative frame buffer
- 2.2. For each frame:
 - Compute pixel difference
 - Append to the cumulative buffer
 - If the buffer reaches the threshold:
 - Segment frames into motion clusters using optical flow analysis
 - Generate anomaly candidate regions

Step 3: Concatenated Adversarial Learning (CAL) for Anomaly Verification

- 3.1. Extract spatial-temporal features from candidate regions
- 3.2. Train adversarial networks (Generator and Discriminator)
 - G learns to generate normal pixel patterns
 - D distinguishes between real and synthetic patterns
- 3.3. Compute anomaly score for each region
 - If the anomaly exceeds the threshold, classify it as an anomaly

Step 4: Anomaly Classification and Output

- 4.1. Apply a pre-trained anomaly classifier to confirm the anomaly type
- 4.2. Store-detected anomalies
- 4.3. Visualize anomaly locations on the UAV display

Return: A (Set of detected anomalies with their spatial coordinates)

the training is evaluated and computed from this segmented frame in the equation below.

$$Seg_a = \frac{o'(mp')}{\sum_{j_{mp}} UA'} * Fr_m + (D_\sigma * Con) + AN' \quad (12)$$

The feature extraction process evaluates the segmentation, where the location is detected from the frame region. The frame region is extracted from the concatenated image, and the mapping is processed to detect anomaly behavior from the crowd image. Here, the segmentation is represented as Seg_a . The segmentation process is described in Fig. 4.

The Seg_a the process is defined using Re_p and Dis' identified between various x_{ip} that is $\partial a'$ evaluated. In this process, the $D_o = reg'$ is the $Seg_a \in (x, y) \neq sm_o$ is the satisfying condition. The Fr_m sequences across various D_σ are reliable to ensure maximum AN' region detection. In this detection process, new Fe_x is not induced for which the x_{ip} that are not similar to LD' output are used to define mc_r segments. If the cumulative $reg' \in D_\sigma$, then Re_p and Dis' are optimally distinguished. Based on the $Sel \forall (x, y) \in Iy'$, the AN' using $(\theta \simeq \omega_n)$ identifies new Dis' , whereas the rest of the regions are used to induce fm_p . Thus, the repeated process is the $TR_\alpha \forall new Fe_x$ whereas $D_o = D_\sigma$ results in false regions that are neglected, and this means there are no variations/ errors in the $Seg_a \forall (Re_p, Dis')$ regions (Fig. 4). The region is segmented into frames, and abnormalities are identified using a CAN learning approach. Here, abnormalities with segmented regions are detected and analyzed below.

$$AL_y = Seg_a(UA') * \sum_{D_\sigma} UA'(reg') + Fr_m * Sel \quad (13)$$

The analysis for the segmented region is performed using the above equation; within the region, pixel variation occurs in both normal and abnormal behaviour. By evaluating the region segmentation, the variation is identified with the concatenated mapping, and the results show whether the extracted feature image is an anomaly or not. By detecting the abnormalities, the maximum dispersion frame is identified.

$$\varnothing_m = UP'(Fr_m) * [(D_\sigma + Iy') * AN'] + \sum_{\partial a'} TR_\alpha(Re_p) + fm_p + D_\mu \quad (14)$$

The dispersion of the image is identified, which is said to be disintegration or particle, and it is described as D_μ . Here, it represents the distributed particle in an image where the proper detection is not done, so it is evaluated with this identification and is labelled as $\varnothing_m$. Therefore, the abnormal object/ human in the image is identified using the maximum dispersion frame. This study aims to investigate the possibility of incorporating Transformer-based designs into the model to make it more resilient and flexible; these designs excel at capturing complex spatiotemporal patterns and long-range relationships. Furthermore, using self-supervised learning, the model can acquire meaningful representations from unlabeled UAV-captured crowd images, thereby reducing the need for massive labeled datasets. With these updates, feature extraction can be enhanced, false positives can be minimized, and the accuracy of anomaly detection in dynamic settings can be further refined. To keep the technique cutting-edge and easily implementable, future research will improve these integrations while keeping computing efficiency for real-time UAV applications. Vision Transformers (ViTs) and Swin Transformers are examples of transformer-based designs that can enhance feature extraction in crowd photos acquired by UAVs by capturing long-range dependencies, which enables more accurate anomaly identification in complex scenarios. Furthermore, contrastive learning and other self-supervised learning (SSL) approaches can make the model more flexible in varied contexts by learning meaningful representations from unlabeled data, thereby reducing reliance on large labeled datasets (e.g., SimCLR, MoCo). In addition, graph neural networks (GNNs) can be used to simulate connections between crowd motions, enhancing the ability to detect subtle irregularities based on spatial-temporal patterns.

4. Results and discussion

The results and discussion are presented in three sections: Ablation, Experimental, and Comparative representations. The system uses the sigmoid linear unit as the activation function, 0.0001 as the learning rate, a batch size of 16, the Adam optimiser, 100 epochs, and a pixel size of 640×640 for the input image, with Xavier uniform initialisation for the weights during input processing. Additionally, the bounding box regression loss is employed to compute the total loss in object detection, and the gradient penalty is utilized in adversarial learning to achieve a feature distribution that is indistinguishable.

4.1. Ablation results

The ablation study presented in this section discusses the true positive rate, confusion matrix representations, and frame update metric analysis. These metrics are used for process verification, as described in Fig. 1. Using multiple iterated instances, the concatenated maps help identify the frame region. Based on the true positives, the D_σ and sm_o are the non-concurrent outputs for Re_p and Dis' . Thus, in the first assessment, the true positives for Fr_m and Seg_a processes are analyzed in Fig. 5.

False rate versus true rate increases for intensity estimation in UAV-captured images. Here, the input is acquired from the crowd image where the intensity features $Iy'(Fe_x + Con) * \sum_{reg} AL_y$ are used to fix the region of interest. Here, the repetition and dissimilar regions are classified $LD'(AN' + D_\sigma) * \prod_{x_{ip}} fm_p$ based on intensity estimation. The computation takes place with features and evaluates the segmentation $Seg_a(reg') + Sel * \sum_{mp'} (Fe_x + UA')$ from the selected region. The evaluation takes place for the detection of anomalies

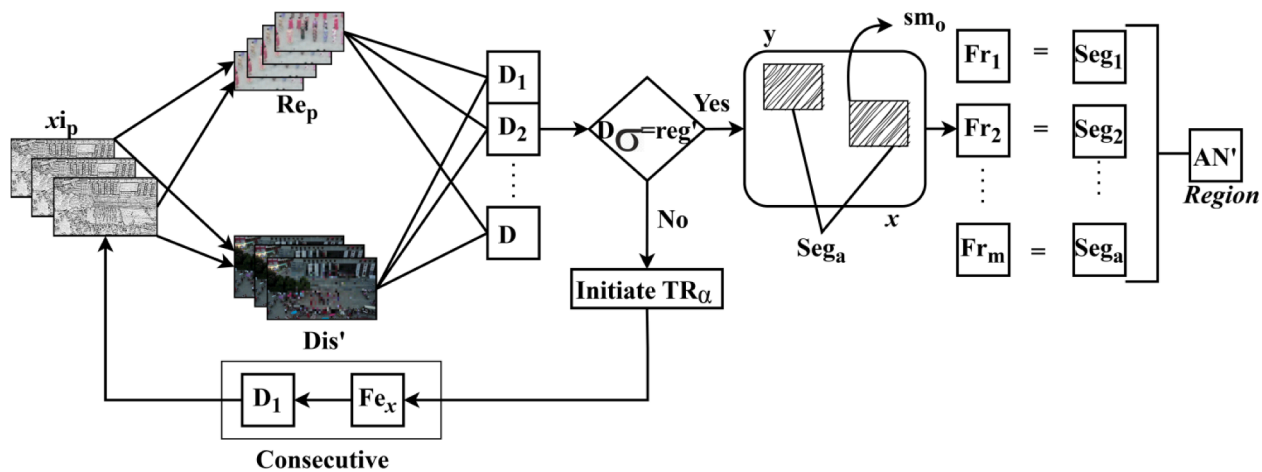
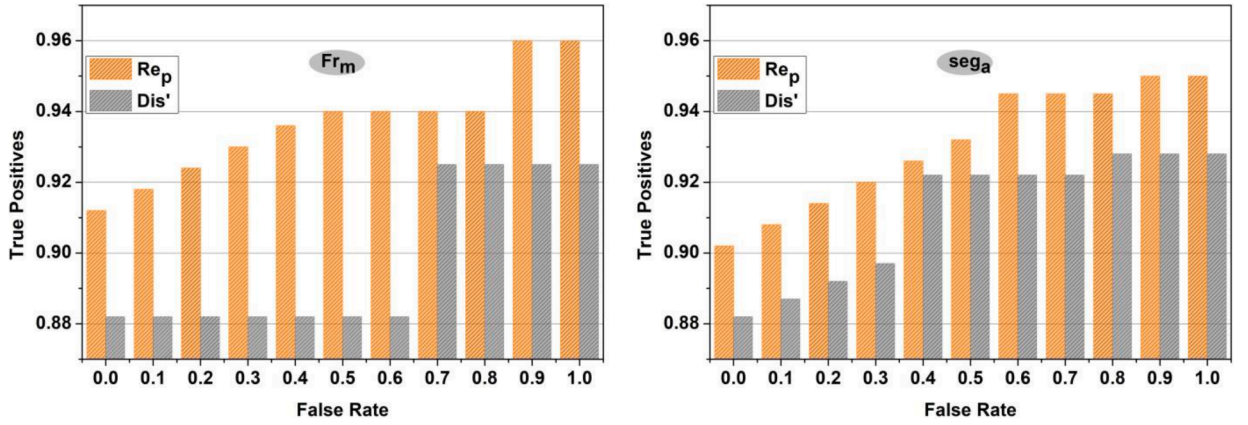


Fig. 4. Segmentation process illustrations.

Fig. 5. True positives analysis for Re_p & Seg_a .

from the gathering and improves the false rate versus the true rate (Fig. 5). The above processes are consolidated in the D_o and D_e improvements analyzed for fm_p using the confusion matrix. The matrix for Re_p and Dis' is presented in Fig. 6.

For classification purposes, the confusion matrix is introduced to evaluate the performance of feature extraction. It is used to classify correct and incorrect images, detect inaccuracies in commonly used data, and rectify them. Using this confusion matrix increases the accuracy and performance by considering four labels: true positive, true negative, false positive, and false negative. Here, the prediction-based result is given by considering the four labels. Based on the labels, the type 1 and type 2 errors are generated for the input image (Fig. 6). Post this matrix assessment, the Frm (Up') for Re_p and Dis' is analyzed in Fig. 7 for the varying iterations.

Training versus frame update increases when repetition occurs during the intensity estimation. The frame region is extracted from the given input $UA' + \prod_{LA'}^{TR_a} (Iy' + Re_p)$ and evaluates the feature extraction process. The computation process is done for the training $(TR_a + Dis') * \prod_{D_e} (fm_p * Iy')$ dissimilar image with CAN learning. The analysis is done for frame update, which is done periodically $Dis'(UA' + sm_0) * Iy' - AL_y$ if there is any dissimilarity, training is given to the UAV image. The frame update is evaluated based on the selected frame as the region of interest, which improves the training (Fig. 7).

4.2. Experimental results

The experimental results are tabulated in this section using the MATLAB outputs and the “VisDrone dataset” (Data, set). This reference comprises 10,209 images derived from over 261,000 frames in 288 drone videos. The images are extensively categorized under various scenarios, objects, humans, vehicles, and more. Most images are annotated with bounding boxes for ease of identification. From the given set of images, the abnormality of the objects/ pedestrians/ vehicles is identified as a change in their position or action that differs from that of other objects of the same kind. For the training, 12,942 images and 639 annotated images are used for testing. MATLAB is installed on a standalone computer with 8 GB of random-access memory and 128 GB of secondary storage. The CPU is an Intel Core i3–2.4 GHz processor and utilises an Intel x86–64 architecture. The GAN architecture is trained for $Frm(UP')$ using 1400 iterations and 4 epochs; the learning rate varies between 0.6 and 1.0 depending on the sm_0 .

The data from the UAV Anomaly Detection (kaggle.com/datasets/luyucwnu/tlmuav-anomaly-detection-datasets) public dataset has been used to evaluate the generality and robustness of the method in various scenarios. Using a software-in-the-loop simulation environment, this dataset identifies UAV anomalies and includes some of the UAV's anomaly logs. A normal state for the UAV is

D_o	(Re_p)					D_σ	(Re_p)					D_σ	(Dis')					D_o	(Dis')				
1	1	0.789	0.824	0.804	0.854	1	1	0.378	0.415	0.456	0.489	1	0.814	0.825	0.886	0.854	1	1	0.598	0.559	0.547	0.478	1
0.8	0.374	1	0.921	0.941	0.932	0.8	0.789	1	0.521	0.556	0.504	0.8	0.924	0.934	0.921	1	0.597	0.8	0.710	0.697	0.628	1	0.912
0.6	0.391	0.521	1	0.928	0.96	0.6	0.821	0.935	1	0.534	0.471	0.6	0.934	0.904	1	0.547	0.628	0.6	0.528	0.605	1	0.889	0.925
0.4	0.581	0.463	0.508	1	0.942	0.4	0.856	0.897	0.925	1	0.529	0.4	0.874	1	0.412	0.569	0.635	0.4	0.415	1	0.836	0.905	0.914
0.2	0.489	0.536	0.417	0.398	1	0.2	0.951	0.960	0.825	0.857	1	0.2	1	0.536	0.654	0.704	0.710	0.2	1	0.815	0.832	0.925	0.940
	0.2	0.4	0.6	0.8	1		0.2	0.4	0.6	0.8	1		0.2	0.4	0.6	0.8	1		0.2	0.4	0.6	0.8	1
	$L\Delta'$						$L\Delta'$						$L\Delta'$						$L\Delta'$				

Fig. 6. Confusion matrix for $(D_\sigma, D_o) \forall (Re_p, Dis')$.

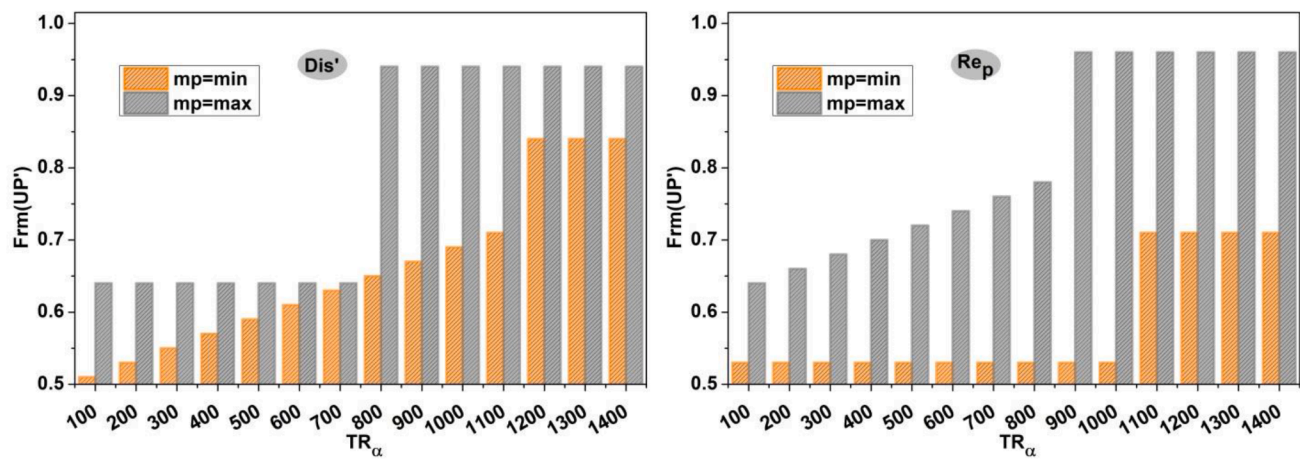


Fig. 7. $Frm(UP')$ analysis $\forall (Dis', Re_p)$ under TR_α .

indicated by Label 0, an anomaly for its GPS is indicated by Label 1, an anomaly for its accelerometer is represented by Label 2, an anomaly for its engine is represented by Label 3, and an anomaly for its RC is represented by Label 4.

Thus, based on these parameters, the experimental outputs are tabulated in Tables 2 and 3.

4.3. Comparison results

The comparison results are discussed using detection accuracy, segmentation precision, sensitivity, specificity, and segmentation error metrics. The impact of the feature extraction rate (0.1 to 1.0) and the frame count (1 to 14) is studied in this section of the comparison results. Besides, to verify the consistency of the proposed method, the existing DeepSDAE (Yang et al., 2023), MC-GCAE (Yan, Xiong & She, 2023), and MSMC—Net (Luo et al., 2024) methods are used in this comparison analysis. The comparisons are presented in Figs. 8 to 10 and in Tables 4 and 5.

4.3.1. Detection accuracy

The detection accuracy increases if intensity features are segmented properly, where $Iy' + Fe_x * (Dis' + TR_a)$ the training is given to the dissimilar image. The accuracy is detected by selecting the image based on the intensity estimation $\sum_{xip}^{Sel} UA' + (mc_r * Re_p)$ and ensures the image is from the crowd. The concatenated mapping is done for the accurate detection of anomaly behaviour $AN'(D_\sigma + Fe_x) * xip$, for all $Fe_x < UA'(reg')$ where the region-based segmentation is processed. The computation is done with the frame update, and then the segmentation is processed $Seg_a(UA' * Iy')$, for all $Seg_a \sim Iy'(sm_0)$, which is based on intensity estimation. Here, region-based segmentation is performed by extracting similar pixels from the input image, ensuring higher detection accuracy.

4.3.2. Segmentation precision

The segmentation precision is enhanced where the anomaly detection is done for the pixel where the mapping $mp'(o' - w_n) * sm_0$, where $mp' \in UA'$ is done with the old and new pixels and ensures the region. The evaluation is done for the concatenated pixel, where it includes the frame region $reg'(Seg_a + UA')$, for all $UA' > Sel$ where the pixel selection is processed. The selection of pixel $Sel(xip + UA') * mp'(Iy')$, which is based on intensity and evaluates the classification. The classification is processed for the repetition and dissimilar image $Re_p + (UA' * mp') * Seg_a$, for all $Seg_a \in xip$ and evaluates the pixel-based segmentation from the extracted region. The region of interest is acquired where $reg'(Fe_x * AL_y) + \prod_{D_\sigma} AN', \forall Iy'(UA')$ and improves the segmentation precision (Fig. 9).

4.3.3. Sensitivity analysis

The sensitivity is reduced if the concatenated mapping is done for the old and new images by considering the intensity $Iy'(mp') - o'$, where $AN' \in o'$ and results in the detection of an anomaly. The anomaly is detected for the frame region where the segmentation is processed for the repetition and dissimilar image $TR_a + Dis'(Iy' + Fe_x) * \sum_o AN'$ and gives training for dissimilar regions. Here, the detection is carried out concerning the feature extraction $Fe_x + (UA' * D_\sigma) + \sum_{AN'} xip \approx AL_y$ and processes the frame region analysis to reduce the sensitivity. If the sensitivity is reduced, then the intensity features are estimated with the concatenated image $Con(UA' * AN') + D_\sigma$, for all $D_\sigma \in AN'$ and performs the mapping with the old image. The mapping is performed to address sensitivity and reduce it by identifying abnormalities (Table 4).

4.3.4. Specificity analysis

The specificity is improved on the crowd image, where the abnormalities are identified from the classification $L\Delta'(Re_p + Dis') * \sum_{Fe_x} UA' + (AL_y)$, for all $AL_y \in Con$ and evaluates the concatenated mapping. From the mapping, the frame region is extracted $Fe_x(Fr_m + reg') * \{[(AL_y + D_\sigma)] * Sel(reg')\} > Iy'(sm_0)$ and ensures a higher specificity for the acquired image. The crowd detection is based on the intensity estimation, where repetition and dissimilar regions are segmented $Seg_a(Re_p + Dis') - Iy'$, where $Fe_x \equiv Sel(UA')$ and process the image selection. Here, the unsafe behaviour is detected from the crowd scenes $UA'(AN') + D_\sigma$, where $D_\sigma \in Con$ and evaluates the detection of an anomaly. From this computation step, the specificity is detected and enhanced (Table 5).

4.3.5. Segmentation error

The segmentation error is reduced if the proper frame region is extracted from the features $Fe_x(UA') + Seg_a(reg')$, where $Seg_a \equiv AN'(D_\sigma)$ here, the intensity-based estimation is evaluated. This error is detected due to the crowded image and evaluates the frame-based region segmentation $Seg_a(reg' + Fe_x) * \sum_{Iy'} sm_0(Fr_m) < xip$ is done for the region-based pixel distribution. The pixel for every region is taken into consideration and provides the lesser segmentation error $Seg_a(UA')$ because only the necessary part in the frame is segmented $Fe_x(Fr_m + reg') * Seg_a$, where $Fr_m \forall D_\sigma$. Therefore, the unwanted region is not segmented due to the feature extraction using CAN learning, which is how the error is reduced when detecting anomalous behavior in the crowd (Fig. 10). The comparison analysis summarizes the following findings, ensuring the proposed method's efficiency over existing approaches. The proposed method increased detection accuracy by 12.81 %, segmentation precision by 14.32 %, sensitivity by 11.8 %, and specificity by 12.41 %, resulting in an 11.38 % reduction in segmentation error. This improvement is observed at the maximum feature extraction rate, i.e., 1.0. The proposed method increased detection accuracy by 13.46 %, segmentation precision by 14.08 %, sensitivity by 12.7 %, and specificity by 12.88 %, resulting in an 11.92 % reduction in segmentation error, as observed for the maximum frame count, i.e., 14.

Table 2
Experimental Outputs for Dis' .

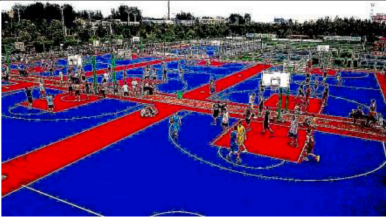
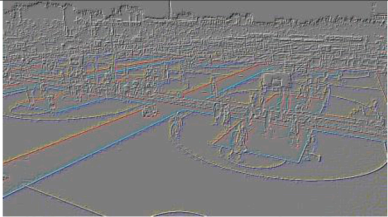
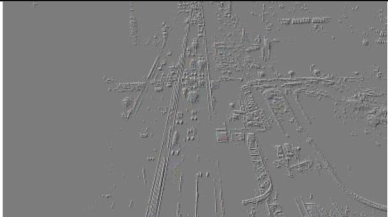
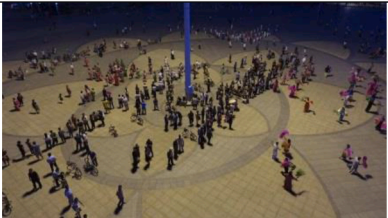
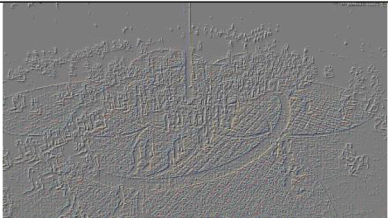
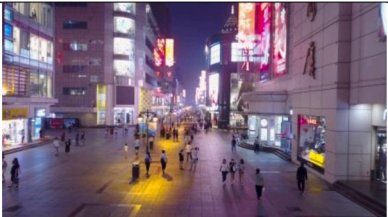
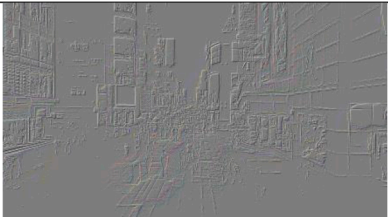
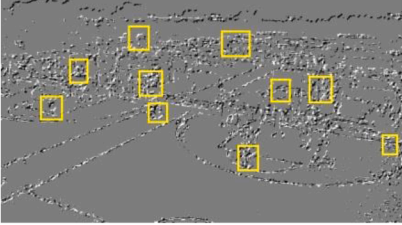
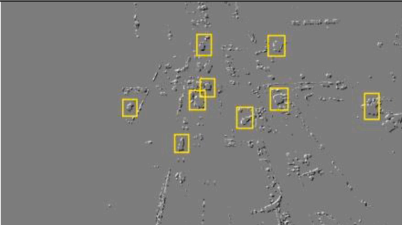
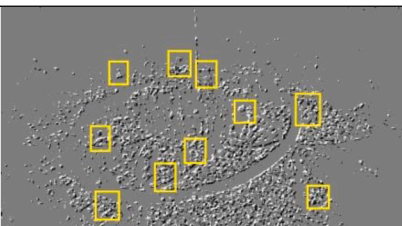
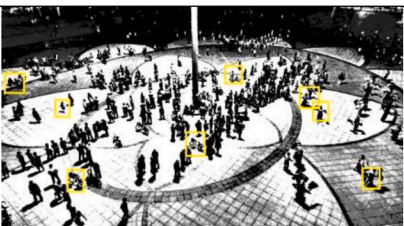
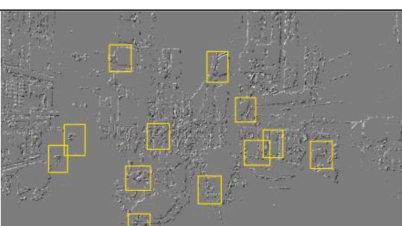
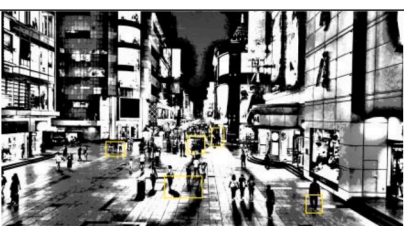
Input Image	Rep	Dis'
		
		
		
		

Table 3
Experimental Outputs for Seg_a .

D_o	D_μ	Seg_a
		
		
		
		

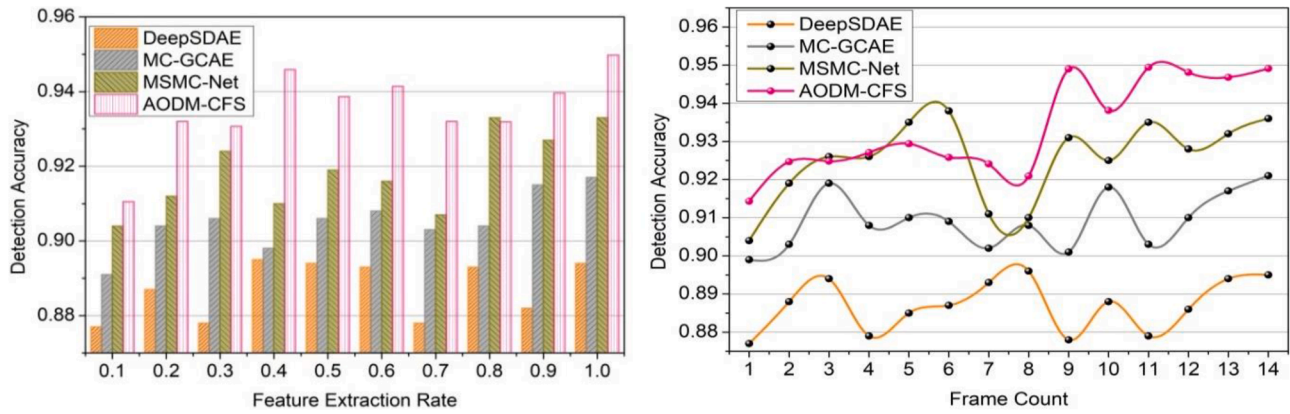


Fig. 8. Detection Accuracy.

Table 4
Sensitivity Analysis.

Feature Extraction	DeepSDAE	MC-GCAE	MSMC—Net	AODM-CFS	Frame Count	DeepSDAE	MC-GCAE	MSMC—Net	AODM-CFS
0.1	0.884	0.902	0.911	0.9246	5	0.907	0.92	0.941	0.9348
0.2	0.891	0.908	0.923	0.9328	6	0.895	0.918	0.939	0.9472
0.3	0.901	0.908	0.91	0.9575	7	0.902	0.927	0.915	0.9381
0.4	0.886	0.898	0.916	0.9619	8	0.892	0.918	0.915	0.9437
0.5	0.906	0.918	0.932	0.9335	9	0.901	0.909	0.939	0.9442
0.6	0.905	0.897	0.942	0.9579	10	0.888	0.923	0.935	0.9453
0.7	0.905	0.922	0.936	0.9536	11	0.888	0.909	0.932	0.9519
0.8	0.887	0.913	0.937	0.9595	12	0.896	0.926	0.932	0.9507
0.9	0.892	0.918	0.934	0.9559	13	0.904	0.916	0.937	0.9504
1	0.901	0.923	0.943	0.9623	14	0.905	0.925	0.945	0.9618

Table 5
Specificity Analysis.

Feature Extraction	DeepSDAE	MC-GCAE	MSMC—Net	AODM-CFS	Frame Count	DeepSDAE	MC-GCAE	MSMC—Net	AODM-CFS
0.1	0.884	0.905	0.918	0.9298	5	0.893	0.917	0.91	0.9517
0.2	0.884	0.904	0.928	0.9329	6	0.885	0.909	0.935	0.9524
0.3	0.889	0.908	0.919	0.9487	7	0.891	0.919	0.937	0.9553
0.4	0.899	0.912	0.921	0.9495	8	0.884	0.911	0.915	0.9325
0.5	0.897	0.919	0.915	0.937	9	0.899	0.902	0.942	0.9582
0.6	0.895	0.915	0.932	0.9463	10	0.893	0.904	0.936	0.9403
0.7	0.896	0.916	0.932	0.9569	11	0.89	0.903	0.934	0.9589
0.8	0.899	0.905	0.942	0.9578	12	0.893	0.908	0.944	0.9314
0.9	0.898	0.912	0.938	0.9603	13	0.889	0.92	0.931	0.9423
1	0.904	0.922	0.948	0.9622	14	0.896	0.921	0.941	0.9549

Theoretically, AODM-CFS adds value by making anomaly detection more resilient in noisy and ever-changing settings by gradually integrating temporal information across frames, which improves object recognition accuracy. In contrast to prior work, this approach addresses the issue of sparse anomalous occurrences that conventional frame-by-frame object identification methods may overlook, as they do not account for the context of minor abnormalities in crowded or moving crowds.

In practice, adversarial learning facilitates the distinction between normal and abnormal objects by training the model on various simulated disturbances, ensuring the detector can handle standard lighting, angle, and occlusion variations seen in UAV imagery. With fewer false positives and the ability to detect irregularities in situations with partial occlusion or fast motion, the findings demonstrate that detection accuracy is significantly improved compared to traditional methods. As these data demonstrate, the method's capacity to identify abnormalities in complex, high-density situations is crucial in real-time applications such as security monitoring, emergency response, and crowd surveillance. In addition, the efficiency of the AODM-CFS is compared with that of existing methods described in references [(Khosravi et al., 2023; Ankit, Santosh & Bhasker, 2024), and (Hussain et al., 2024)], and the obtained results are presented in Table 6.

The results show that the proposed AODM-CFS framework outperforms current techniques. Its accuracy of 0.96 outperforms TDS-Net (0.94), Khosravi et al.'s emotion prediction model (0.91), and PeopleNet (0.88), demonstrating its ability to distinguish between normal and abnormal crowd behaviors. Precision (0.94) and recall (0.95) demonstrate a balanced ability to minimize false positives and accurately capture actual anomalies in high-density or obstructed situations. The F1-score of 0.945 shows that the model performs consistently under different settings. Due to its integrated object-temporal detection pipeline and adversarial training technique, the model can generalize more effectively in complex surveillance settings, such as UAV-based crowd monitoring. Previous works either lack temporal awareness, struggle with motion and occlusion, or rely heavily on static features, limiting their applicability in dynamic real-world settings.

The AODM-CFS framework, which has been developed, makes a theoretical contribution by enhancing the resilience of object recognition in complex and dynamic situations, which is accomplished by strengthening anomaly detection through the progressive integration of temporal context across video frames for each frame. Our method captures the spatial-temporal evolution of objects, enabling the identification of sparse or contextually modest abnormal events that contrast with standard frame-by-frame detection models, which frequently fail to detect subtle or temporary anomalies in high-density crowds. A fundamental weakness of previous models is that they overlook temporal dependencies, which is especially problematic in the context of unmanned aerial vehicle (UAV) surveillance, as the appearance of objects can vary rapidly due to factors such as motion, perspective, and partial occlusion. When it comes to the practical side of things, the addition of adversarial learning during training enables the model to distinguish between normal and abnormal patterns more efficiently, even when subjected to disruptions that are commonly encountered in the real world, such as variable lighting, angles, and occlusions. In real-time situations, this results in a substantially lower number of false positives and a higher detection accuracy. These enhanced detection capabilities are of direct advantage to applications such as public safety surveillance, monitoring crowd behaviour, and emergency response coordination, all of which require accurate and rapid identification of anomalies. As a result, the AODM-CFS not only contributes to the advancement of theoretical modelling of crowd anomaly

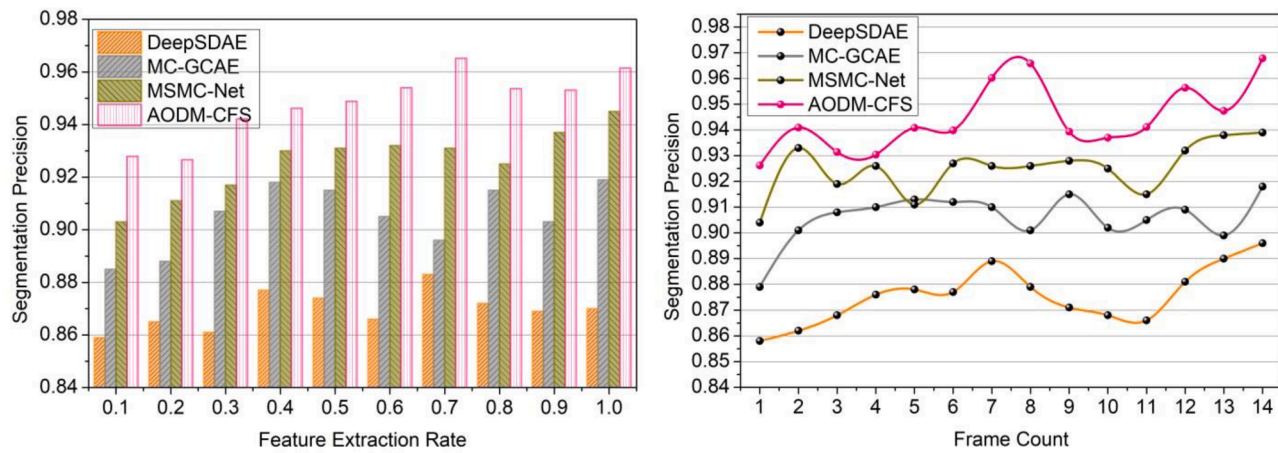


Fig. 9. Segmentation precision.

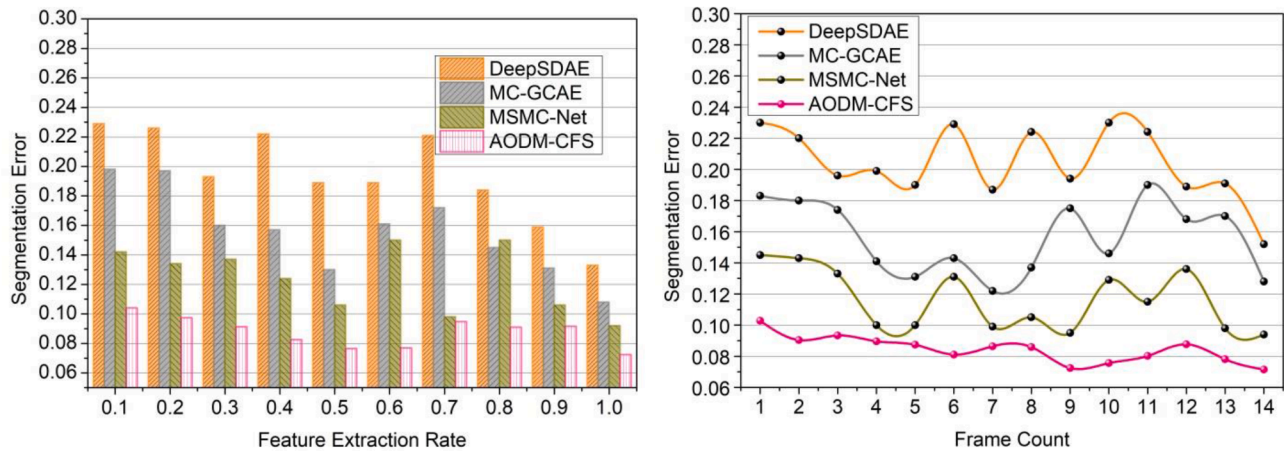


Fig. 10. Segmentation error.

Table 6
Efficiency analysis of AODM-CFS.

Method	Accuracy	Precision	Recall	F1-Score
Modified Transfer Learning + Fuzzy Logic (MTL-FC) (Khosravi et al., 2023)	0.914	0.895	0.885	0.885
Head-Mounted Counting with Temporal Aggregation (HMC-TA) (Ankit, Santosh & Bhasker, 2024)	0.883	0.843	0.856	0.845
Transformer Dual-Stream for Anomaly Detection (TDS-Net) (Hussain et al., 2024)	0.945	0.922	0.937	0.925
AODM-CFS	0.962	0.944	0.953	0.945

detection but also demonstrates that it is a solution that is both scalable and resilient for applied UAV-based monitoring systems. AODM-CFS has good anomaly detection accuracy; however, real-time UAV systems present various practical obstacles. Because UAVs have limited computing power, computational efficiency must be considered first. To reduce computational effort, consider pruning and quantizing models or utilizing TinyML platforms. Second, the power limits of UAV hardware necessitate energy-efficient inference, which can be achieved by utilising low-power AI accelerators or offloading computationally intensive calculations to edge servers. Ultimately, anomaly detection latency is crucial for informed decision-making in security and emergency response. AODM-CFS's modular design enables partial inference on-device with real-time outputs and asynchronous full-frame analysis, providing speed and accuracy. These factors underpin future research on the integration of UAV-based surveillance.

5. Conclusion

This article proposes an anomaly object detection method using cumulative frame segmentation. The proposed method extracts intensity features over different pixel variations and arrangements. The extracted features are used to segment frames by computing the maximum diversions. Using these diversions, the pixel repetitions and dissimilar feature variations are calculated. The concatenated adversarial learning identifies the overlapping variations above to compute independent regions. Depending on the concatenation maps generated, the overlapping regions are segmented. This segmentation is pursued to train the network map to detect feature diversions. In this process, the abnormal region is segmented using concatenated selections. Thus, the increasing iterations verify the anomaly region detection by reducing false rates and mean error. The proposed method increased detection accuracy by 12.81 %, segmentation precision by 14.32 %, sensitivity by 11.8 %, and specificity by 12.41 %, resulting in an 11.38 % reduction in segmentation error, which is observed at the maximum feature extraction rate. In the future, concurrent layered assessments are to be made for the Dis' and Re_p classified regions, which would eventually reduce the converged replications in selecting reg' by transforming the layers from smaller to higher resolution. This replication assessment is recommended to be performed using transformer learning. In this assessment, the layered neural network is to be gradually increased to reduce convergence complexities. In practical applications, a significant barrier is the need for computing resources, as UAVs often possess limited processing capabilities, which complicates the real-time deployment of intricate, deep learning models. This study suggests utilising model compression approaches, including quantisation and pruning, to minimise computational costs while maintaining high detection accuracy. Moreover, edge computing and streamlined neural network topologies may improve real-time processing capacities, facilitating effective onboard anomaly detection. A significant challenge is identifying anomalies in real-time while processing high-resolution UAV video feeds. Conventional models may experience latency issues due to the substantial influx of data. This study proposes adaptive frame segmentation algorithms that dynamically modify processing according to crowd density and movement patterns, prioritizing essential frames for anomaly detection. In addition, sensor noise that influences image quality, regulatory constraints that restrict flying operations, and concerns about the scalability of large-area deployment are potential implementation challenges. Furthermore, effective data transmission techniques, including improved wireless communication protocols, may improve responsiveness in real-time monitoring situations.

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CRedit authorship contribution statement

Fangfang Ye: Writing – original draft, Validation, Methodology, Conceptualization. **Jinming Wang:** Writing – original draft, Validation, Project administration, Methodology. **Cao Shuhua:** Writing – review & editing, Project administration, Formal analysis, Conceptualization. **Zhou Dong:** Writing – review & editing, Writing – original draft, Resources. **Ting Wang:** Writing – original draft, Methodology, Investigation, Formal analysis. **Ezzeddine Touti:** Writing – original draft, Software, Resources, Formal analysis, Data curation.

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Data availability

Data will be made available on request.

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